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SUPERSEDING
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9 November 1984

MILITARY SPECIFICATION
SCORING SET, MISSILE, NON-COOPERATIVE SYSTEM
AN/DSQ-37(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and, is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the AN/DSQ-37(V) Scoring Set, Missile, Non-Cooperative System, herein referred to as the Scoring Set, for use as a Miss Distance Indicator (MDI) for aerial and surface targets.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5821

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SPECIFICATIONS

MILITARY

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures, Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic Aeronautical and Aeronautical Support Equipment, Including Ground Support Equipment

STANDARDS

FEDERAL

FED-STD-595	Color

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461B	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Measurements of Electromagnetic Interference Characteristics
MIL-STD-704C	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Reliability Program for System Equipment, Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

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MIL-STD-810C	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-965	Parts Control Program
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2078	Requests for Preparation of Support Equipment Depot Level Rework Standards
MS 3134	Connector, Receptacle, Electric, Solder Type, Jam Nut Mounting, Push-Pull Coupling, Series 1
MIL-STD-45662	Calibration System Reqirements

2.1.2 Other Government documents, drawings, and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command
(Code Ident. 30003)

1356AS3021	Scoring Set, Missile Non-Cooperative System AN/DSQ-37(V)

PUBLICATIONS

Inter-Range Instrumentation Group (IRIG)

IRIG 106-80	Telemetry Standards

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specified acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.1.1 First article sample. The first article sample shall be as specified in the contract or purchase order (see 6.2.1). The first article sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production.

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3.2 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic technology shall be considered in accordance with microelectronic requirements specified herein.

b. Other parts and materials requirements shall be in accordance with MIL-E-5400.

c. Nonrepairable subassemblies shall be used in accordance with MIL-E-5400.

d. Parts shall be selected using the following order of preference:

(1) Parts listed on a Qualified Parts List (QPL) with two or more sources.

(2) Parts listed on a QPL with only one source.

(3) Commercial parts with two or more sources.

(4) Commercial parts with only one source.

3.2.1 Standard parts. Parts specified in 3.2(d)(1) are standard parts and do not require approval from the procuring activity. Parts selected for use from 3.2(d)(2), (3) and (4) are considered nonstandard and shall require approval by the procuring activity. Nonstandard parts and approval requests shall be prepared using DD Form 2052 in accordance with the contract or purchase order (see 6.2.2).

3.2.2 Commercial parts. Commercial parts specified in 3.2(d)(3) and (4) shall be readily available, nationally catalogued, off-the-shelf items.

3.2.3 Development contractor parts. Parts manufactured by the development contractor only, will not be accepted without full design disclosure including any special processing required to duplicate the part.

3.2.4 Proprietary parts. Proprietary parts, assemblies or processes shall not be used.

3.2.5 Nonstandard parts. Use of nonstandard parts in first article or production units prior to Government approval shall be at the sole risk of the contractor and shall be subject to replacement if the Government does not approve proposed nonstandard parts.

3.3 Scoring Set description. The Scoring Set shall be a non-cooperative scalar scoring system for missile and gunnery weapon systems evaluation and training operations. The Scoring Set shall be comprised of the following subsystems:

a. An MDI Sensor

b. A Voltage Controlled Oscillator (VCO)

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c. A commercially available Telemetry (TM) Transmitter

d. TM Transmitter Bandpass Filter or functional equal

3.3.1 Scoring Set function. The Scoring Set functional block diagram is shown in Figure 1.

3.3.2 Interface description. The Scoring Set shall interface with target systems by associated installation kits which contain sensor and telemetry antennas, mechanical brackets and radio frequency (RF) cabling peculiar to each target system. The Scoring Set shall also interface and be compatible with the following associated equipment.

Nomenclature	Designation/part number
Doppler Simulator	Babcock BTS-83 or equal
Test Set	Range Simulation System as described in Figure 2

3.4 Characteristics.

3.4.1 Performance characteristics. The Scoring Set consisting of the subsystems in 3.3 shall conform to the requirements as specified herein.

3.4.1.1 Scoring Set sensor.

3.4.1.1.1 Range and accuracy. The Scoring Set shall have a scoring range of 75 feet with a one-tenth (0.1) square meter target and with Scoring Set sensor output jack (J2) to MDI Sensor antenna cable length from three feet up to and including 12.5 feet of RG-142 B/U cable. The accuracy shall be ±1 foot.

3.4.1.1.2 Closing velocity. The Scoring Set shall score projectiles when the projectile to Scoring Set closing velocity is not less than 200 and not greater than 5,000 feet per second, inclusive.

3.4.1.1.3 Cable lengths. The Scoring Set shall meet all requirements specified herein with any Scoring Set MDI Sensor antenna jack (J2) to MDI sensor antenna cable lengths of RG-142 B/U from three feet to 12.5 feet, inclusive. There shall be only external potentiometer (pot) adjustment(s) to compensate the Scoring Set for any cable length in the range specified above. The Scoring Set shall be set for a 12.5 foot cable length. The external pot adjustment shall be adequate to compensate the Scoring Set for a 15.5 foot cable length.

3.4.1.1.4 MDI Sensor antenna. The Scoring Set shall operate with a single MDI Sensor transmit-receive antenna. The required system performance as specified herein shall be obtained with an omnidirectional antenna at the MDI Sensor frequency with a gain of -3.0 decibels isotropic (dBi) and a voltage standing wave ration (VSWR) of less than or equal to 1.5:1 (50 ohms reference).

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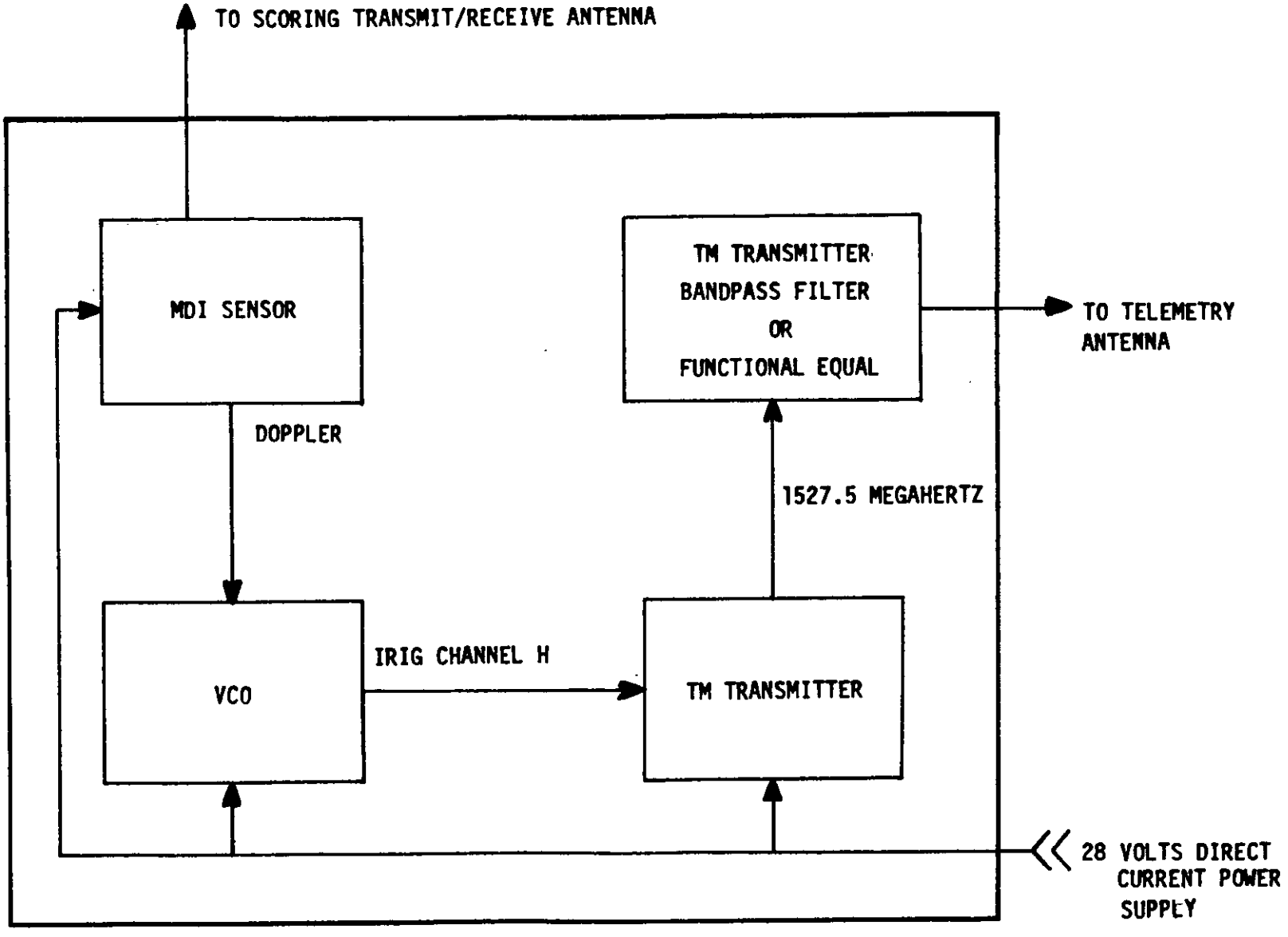
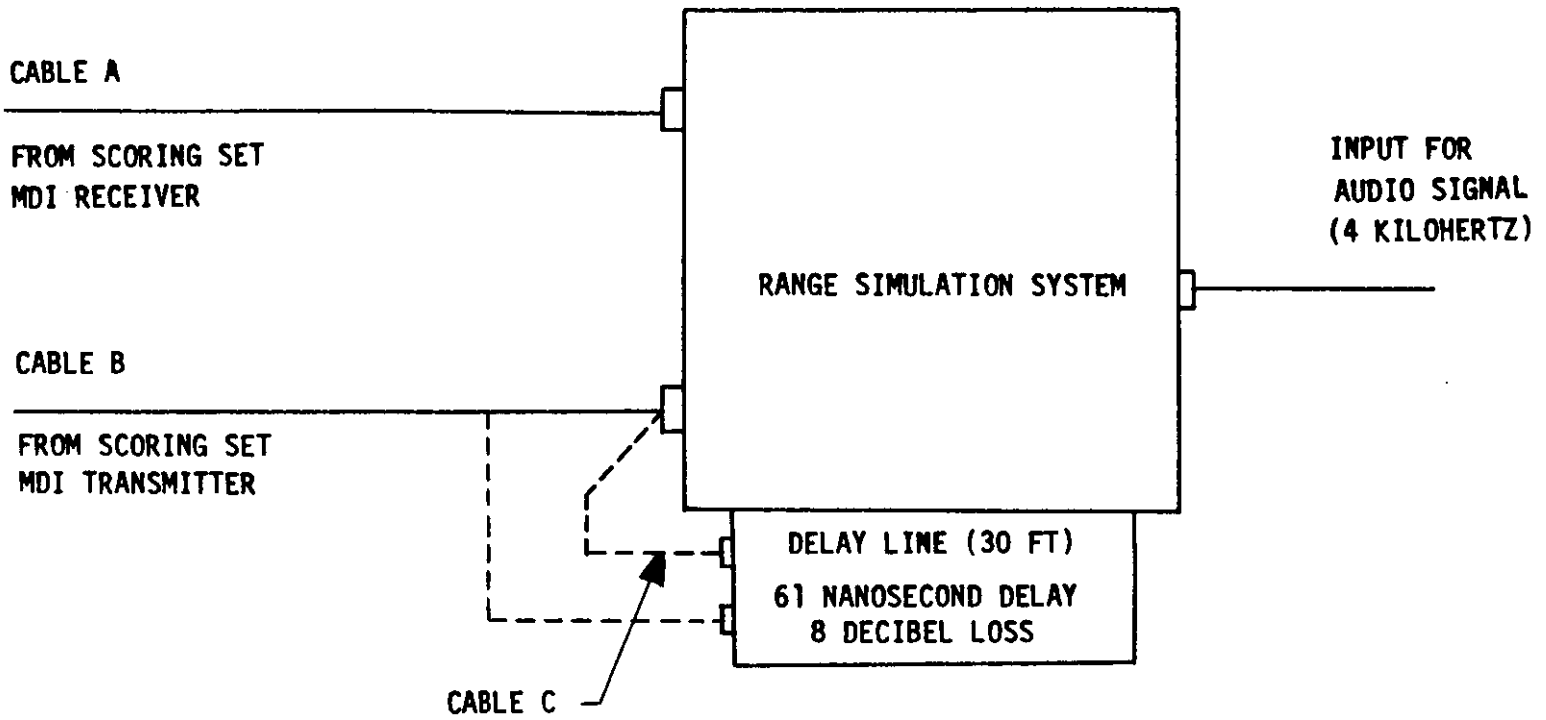


FIGURE 1. Scoring Set AN/DSQ-37(V) functional block diagram.

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NOTES:

1. CABLE A = CABLE B = FLIGHT LENGTH CABLE.
2. FOR RANGE 50 FEET (FT) TO 80 FT, DISCONNECT CABLE B FROM RANGE SIMULATOR AND CONNECT TO DELAY LINE. CONNECT CABLE C BETWEEN TRANSMITTER PORT OF RANGE SIMULATOR AND DELAY LINE (DOTTED LINES).
3. CAPABLE OF MEASURING LOOP SENSITIVITY FOR RANGES 10 TO 80 FT IN 2.5 FT INCREMENTS.

FIGURE 2. Range Simulation System.

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3.4.1.1.5 Frequency. The MDI Sensor shall operate at 1775 ±5 Megahertz (MHz).

3.4.1.1.6 Transmitter peak power. The radiated power outputs of the MDI Sensor shall be not less than three watts peak under all combinations of operating power and operating temperatures (see 3.4.2 and 3.5.1.2).

3.4.1.1.7 Transmitter pulse characteristics. The MDI Sensor pulse width measured at 50 percent peak deflection shall be 100 ±10 nanoseconds. The pulse rise-time shall be not greater than ten nanoseconds and the pulse fall-time shall be not greater than ten nanoseconds between the ten and 90 percent points. The pulse shall be flat plus or minus a tilt of ten percent of the peak voltage. All pulse characteristics shall be measured with the same sampling scope vertical gain.

3.4.1.1.8 Transmitter pulse repetition frequency (PRF). The PRF of the MDI Sensor shall be 500 ±50 Kilohertz (kHz).

3.4.1.1.9 Receiver sensitivity. The MDI Sensor shall have a receiver sensitivity of -93 decibels-milliwatt (dBm) or greater than -93 dBm in the negative direction for a six decibel video signal plus noise-to-noise ratio.

3.4.1.1.10 Receiver video output noise level. The MDI Sensor video output noise level shall be -32 ±2 dBm as measured by a root mean square (rms) voltmeter (zero dBm = one milliwatt (mW) into 600 ohms).

3.4.1.1.11 Receiver video (doppler) output signal. The doppler output shall have a frequency response range of 1150 ±50 Hertz (Hz) to 18045 ±785 Hz (three decibel (dB) down points from response at four kHz). The response at 400 ±20 Hz shall be not less than 20 dB down from the four kHz reference level. The maximum doppler output amplitude shall be 4.5 ±0.5 volts peak-to-peak (Vpp). The receiver video output signal shall be capable of driving a 50-ohm load without signal degradation.

3.4.1.1.12 Receiver sample and hold pulse. The MDI Sensor receiver sample and hold pulse width shall be 30 ±3 nanoseconds and the delay from receiver turn-on to the sample and hold pulse shall be 27 ±3 nanoseconds. The sample and hold pulse amplitude shall be below zero volts before its turn-on.

3.4.1.1.13 Loop sensitivity. Loop sensitivity (see 6.5) shall be determined by the path loss method where the transmitted signal is delayed, attenuated, and modulated to simulate a radar return from a one-tenth square meter target at a specified range. All loop sensitivity measurements shall be made by utilizing the Range Simulation System as defined in Figure 2 with a modulation level of four kHz at five Vpp from a 50-ohm source. All loop sensitivity measurements shall be referenced to the MDI Sensor transmit/receive antenna. Losses due to the cable (three feet to 12.5 feet) used between the MDI Sensor antenna jack (J2) and the MDI Sensor antenna shall not be included in the loop sensitivity computations. The sensitivity for

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different radar ranges with all cable lengths of RG-142 B/U specified in 3.4.1.1.3 shall be as follows:

a. The loop sensitivity from 35 to 70 feet shall be not less than 125 dB.

b. The loop sensitivity at 75 feet shall be 122 ±1 dB.

c. The loop sensitivity at 77.5 feet shall be not greater than 119 dB.

d. The loop sensitivity at 80.0 feet shall be not greater than 111 dB.

e. The loop sensitivity from ten to 25 feet shall be monotonically increasing. (See 6.4 for definition of a monotonically increasing function.)

f. The loop sensitivity at ten feet shall be not greater than 93 dB.

g. The loop sensitivity at 25 feet shall be 115 ±5 dB.

3.4.1.1.14 Desensitized "dead" zone. There shall be an adjustable desensitized zone extending from the MDI Sensor antenna out to not less than 30 feet to minimize the effects of reflections from the carrying vehicle. The desensitized zone shall be set as described in 3.4.1.1.13 e, f, and g for all cable lengths of RG-142 B/U from three feet to 12.5 feet, inclusive.

3.4.1.1.15 TM Transmitter frequency rejection. The receiver shall not produce video (doppler) outputs greater than three dB above the receiver video output noise level with 30 dB external isolation simultaneously applied between the Scoring Set TM Transmitter antenna output jack (J3) and the MDI Sensor antenna jack (J2) and between another two watt L-band TM Transmitter (different frequency) with ±250 kHz deviation and the MDI Sensor antenna jack (J2). This continuous TM rejection frequency range shall be 1527.5 ±5 MHz without adjustment of the notch filter.

3.4.1.1.16 Internally-generated non-scoring signals. Internally-generated non-scoring signals shall be not greater than the following levels when the MDI Sensor antenna cable is terminated with 50 ohms and 1.5:1.0 VSWR loads:

a. "Beat notes": Three dB above the reference receiver video output noise level (see 3.4.1.1.10)."

b. Discrete spectral components: Ten dB above the reference receiver video output noise level as measured on a spectrum analyzer using a resolution bandwidth setting of 300 Hz, a video bandwidth of ten Hz and a frequency bandwidth of zero to 100 kHz.

c. Random-noise level shifts: ±2 dB about the reference receiver video output noise level.

3.4.1.1.17 Microphonics. The Scoring Set shall not be microphonic when tested in accordance with 4.5.1.5.

3.4.1.1.18 Open and short circuit loads. The MDI Sensor shall operate without degradation in performance or damage to the unit after being subjected to open and short circuit loads to the MDI Sensor antenna jack (J2).

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3.4.1.1.19 Cable phase sensitivity. The MDI Sensor shall not be sensitive to any phase changes of the MDI Sensor antenna jack (J2) cable between three and 12.5 feet with 50 ohms and 1.5:1.0 VSWR loads (see 3.4.1.1.16 for non-scoring signal levels).

3.4.1.1.20 Internal timing reference. The timing reference for all pulses and delays shall be the trailing edge of the MDI Sensor transmit pulse.

3.4.1.1.21 External adjustments. The following external adjustments to the Scoring Set shall be provided if these parameters are variable:

a. VCO ouput voltage.

b. Sample and hold pulse width.

c. Sample and hold delay.

d. Sensor transmitter pulse width.

e. Sensor receiver turn-on delay.

3.4.1.2 Scoring Set Voltage Controlled Oscillator. The VCO shall be modulated by the receiver video output and the VCO shall modulate a frequency modulation (FM) TM Transmitter (see 3.4.1.3).

3.4.1.2.1 Frequency. The VCO shall operate at IRIG Channel H. The center frequency shall be 165 ±3.3 kHz.

3.4.1.2.2 VCO deviation. The clipper doppler signals shall deviate the VCO to 140.25 kHz (-0, +5 kHz) and 189.75 kHz (-5, +0 kHz). The VCO deviation shall be linear to +2 percent, using 11 points to check the linearity.

3.4.1.2.3 Output voltage. The VCO output shall be externally adjustable to deviate the TM Transmitter (see 3.4.1.3) from ±100 to ±300 kHz. The VCO shall be set for ±250 kHz TM Transmitter deviation. There shall be a ±10 percent tolerance on the 250 kHz deviation.

3.4.1.3 Scoring Set TM Transmitter. The TM Transmitter shall be modulated by the VCO. The TM Transmitter shall be removable from the Scoring Set for independent testing of the requirements specified in 3.4.1.3.1 through 3.4.1.3.14. The TM Transmitter shall meet all requirements specified herein when terminated with any load up to and including 2.0:1.0 mismatch at any phase angle.

3.4.1.3.1 Frequency. The frequency shall be 1527.5 MHz.

3.4.1.3.2 Frequency stability. Frequency stability shall be ±0.003 percent of 1527.5 MHz.

3.4.1.3.3 Spurious emissions. Spurious emissions shall conform to IRIG 106-80 including in channel frequencies (all frequencies between 1526.5 MHz and 1528.5 MHz).

3.4.1.3.4 Type of modulation. Modulation type shall be FM.

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3.4.1.3.5 Power output. The transmitter output power measured at the transmitter antenna jack (J3) shall be not less than two watts when terminated with any load (resistive and reactive) having a VSWR of up to 2.0 to 1.0 with respect to 50 ohms under any combination of operating power and environmental conditions specified herein, e.g., 24 volts direct current (VDC) and -40 to +71 degrees Centigrade (°C).

3.4.1.3.6 Maximum carrier deviation. Maximum carrier deviation shall be not less than ±500 kHz.

3.4.1.3.7 Deviation sensitivity. Deviation sensitivity shall be ±300 kHz (peak) per 1.0 volt root mean square (Vrms) (sine wave).

3.4.1.3.8 Incidental FM. The incidental FM measured over a bandwidth from ten Hz to 300 kHz shall be limited to two kHz rms under any combination of operating power and environmental conditions specified herein.

3.4.1.3.9 Modulation frequency response. The frequency response characteristic of peak-to-peak carrier deviation versus modulation input frequency shall be ±1.5 dB from 100 Hz to 500 kHz.

3.4.1.3.10 Output impedance. Output impedance shall be 50 ohms.

3.4.1.3.11 Open and short circuit loads. The TM Transmitter shall employ an internal isolator and shall operate without degradation in performance and without sustaining damage after being subjected to any load impedence including open and short circuit loads on the RF output.

3.4.1.3.12 Amplitude modulation. Amplitude modulation shall be less than two percent of the carrier rms amplitude under any combination of operating power and environmental conditions specified herein.

3.4.1.3.13 Environmental conditions. The Scoring Set TM Transmitter shall meet all requirements of 3.5 through 3.5.1.2.

3.4.1.3.14 Input current. Input current shall be not greater than 650 milliamperes (ma).

3.4.1.4 Scoring Set TM Transmitter Bandpass Filter (or functional equal). Spurious emissions generated by the TM Transmitter shall not be detected by the MDI Sensor. The bandpass filter shall have the following performance characteristics.

a. Center frequency: 1527.5 MHz

b. Bandwidth: Not less than ±7.5 MHz from 1527.5 MHz at the three dB points

c. Insertion loss: Less than 0.9 dB

d. Attenuation: Greater than 60 dB from 1620 to 2000 MHz

e. Input Power: Not less than 12 watts

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3.4.2 Electrical characteristics.

3.4.2.1 Operating power. The Scoring Set shall meet all applicable requirements of MIL-STD-704 and shall perform as specified herein when energized from the following power sources having characteristics and limits as defined in MIL-STD-704 except as otherwise specified herein. The power required shall be not greater than the amounts specified herein:

a. 28 volts direct current (dc) power ranging from 24 to 32 VDC

b. Input power not greater than 60 watts

c. 1.5 volts (peak) maximum alternating current (ac) ripple from 400 Hz to 2400 Hz

3.4.2.2 Degraded performance. Degraded performance is permissible for all operating power voltage transients from 50 microseconds to 0.5 second duration in accordance with MIL-STD-704 for normal electrical system operation. Operation shall return to a condition of specified performance in accordance with the requirements of this specification, with no resulting damage to the Scoring Set, within 0.5 second after the 50 microseconds to 0.5 second transient expires.

3.4.2.3 Reverse polarity protection. The Scoring Set shall not be damaged by 28 VDC input voltage polarity reversal for 30 seconds.

3.4.2.4 Grounded input. The Scoring Set shall be such that the negative of primary direct current power source may be grounded externally with no adverse effect on the Scoring Set.

3.4.2.5 No load condition. The Scoring Set shall start to operate with the application of an input voltage within the range specified herein at any temperature between -40 and +71°C. Scoring Set starting at ambient temperature (23°C ±10°C) shall be independent of the rate of rise of the applied operating power.

3.4.2.6 Overheating. The Scoring Set shall not overheat while operating for periods not less 24 hours.

3.4.3 Physical characteristics.

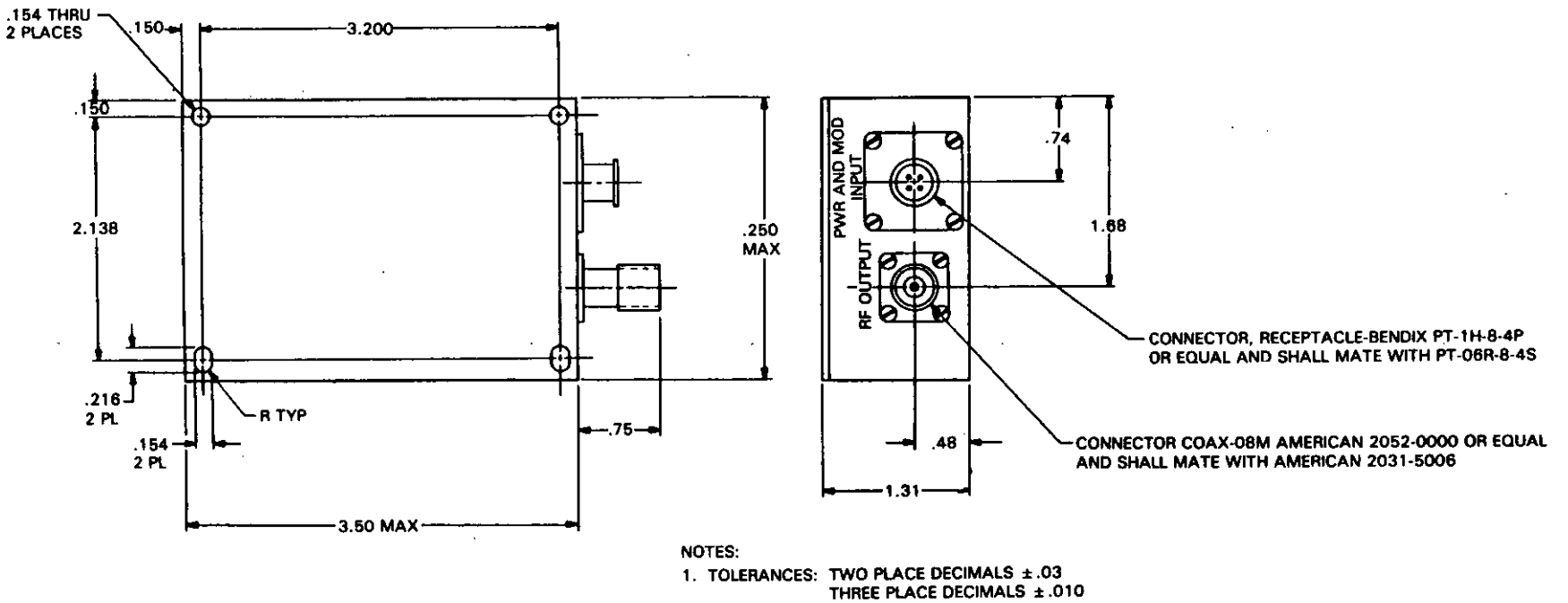
3.4.3.1 Form factor. The Scoring Set shall be in accordance with Drawing 1356AS3021 and as specified herein.

3.4.3.1.1 External circulators. No external circulators and associated cabling shall be utilized.

3.4.3.1.2 Scoring Set baseplate. The dimensions of the baseplate (see 6.6) and the mounting holes shall be in accordance with Drawing 1356AS3021.

3.4.3.1.3 TM Transmitter dimensions. The dimensions of the TM Transmitter and the type of connectors used shall be in accordance with Figure 3.

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NOTES:
1. TOLERANCES: TWO PLACE DECIMALS ±.03
THREE PLACE DECIMALS ±.010

FIGURE 3. Transmitter, telemetry.

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3.4.3.2 Weight. The weight of the Scoring Set shall be not greater than 15 pounds.

3.4.3.3 Scoring Set external connectors. The connectors shall be on one end of the Scoring Set enclosure only and shall be spaced not less than 1.125 inches apart (center-to-center) for easy installation or removal of mating connectors by hand.

3.4.3.3.1 Clearance. Connectors shall not extend beyond the dimensions of the baseplate.

3.4.3.3.2 Power jack (J1). The Scoring Set power jack shall be an MS 3134E3-5OP type connector and shall be uniformly keyed with keyway at 12:00 o'clock (toward top with baseplate down).

3.4.3.3.3 MDI Sensor antenna jack (J2). The MDI Sensor antenna jack shall be a female SMA type connector and shall be rigidly mounted (reinforced) so as to withstand extensive, heavy use.

3.4.3.3.4 TM Transmitter antenna jack (J3). The TM Transmitter antenna jack shall be a female TNC type connector.

3.4.3.3.5 Receiver video output jack (J4). The unattenuated receiver video output (the VCO input) shall be brought out at J4. J4 shall be on the same enclosure face as J1, J2, and J3 and shall be a female BNC type connector.

3.4.3.3.6 Baseplate holes. All connectors, plugs, jacks, and attached cabling shall allow for easy access to, and clearance around, all baseplate mounting holes in accordance with Drawing 1356AS3021.

3.4.3.4 MDI Sensor timing test points. The MDI Sensor transmitter switch timing pulse, receiver switch timing pulse, and the sample and hold pulse shall be made easily accessible when the Scoring Set baseplate or enclosure cover is removed.

3.4.3.5 External adjustments. All external adjustments shall be made by means of screw-sealed, labeled, access holes. The access holes shall be centered over the intended pot adjustments. The access holds shall be large enough for ease of adjustment.

3.4.3.6 Modularity. The Scoring Set shall comprise a modular design. No one module shall exceed the physical dimensions of 8.35 inches by 5.6 inches by 3.4 inches. All modules shall be contained within the overall enclosure specified in 3.4.3.1.

3.4.4 Reliability. The contractor shall conduct a reliability program in accordance with MIL-STD-785. The contractor shall prepare a reliability test and demonstration report (see 6.2.2).

3.4.4.1 Operational stability. The Scoring Set shall operate with performance as specified herein continuously, or intermittently, for a period of not less than 30 days without necessity for readjustment.

3.4.4.2 Operating life. The Scoring Set shall be designed to have a total operating life of not less than 5,000 hours.

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3.4.4.3 Reliability in mean-time-between-failures (MTBF). The Scoring Set shall have a test MTBF of not greater than 100 hours when tested and accepted as specified in 4.7.

3.4.5 Maintainability. The Scoring Set shall conform to the maintainability requirements of MIL-STD-2078 and as specified herein.

3.4.5.1 Mean-time-to-repair (MTTR). The MTTR of the Scoring Set shall be not greater than 60 minutes. The arrangements of parts shall allow ease of maintenance as specified in MIL-E-5400.

3.4.5.2 Access time. The time required to disassemble, dismantle, or open the Scoring Set or otherwise gain access to a Shop Replaceable Assembly (SRA) shall be not greater than ten minutes for 99 percent of all individual SRA replacements.

3.4.5.3 Replacement without adjustment. All SRAs having the same part numbers that are designed for removal and replacement as the normal corrective maintenance action shall be physically and functionally interchangeable with one another between all locations of use without adjustment of either the inserted module or equipment in not less than 98 percent of all instances of replacement and shall conform to the requirements of MIL-STD-454, Requirement 7.

3.5 Environmental conditions. The Scoring Set shall meet the performance requirements of Section 3, unless otherwise specified herein, under any combination of the environmental conditions specified in 3.5.1 through 3.5.11.

3.5.1 Temperature.

3.5.1.1 Non-operating. The Scoring Set shall not be damaged or the operational performance degraded after being exposed to non-operating temperatures between -40 to +71°C for a period of 24 hours.

3.5.1.2 Operating. The Scoring Set shall operate between -40 and +71°C baseplate temperature.

3.5.2 Humidity. The Scoring Set shall perform as specified herein at a relative humidity of 95 percent for both continuous and intermittent periods for the worst case surface humidity temperature relationships specified in MIL-STD-210, including conditions wherein condensation takes place in the form of water or frost.

3.5.3 Watertight integrity. The Scoring Set enclosure shall be watertight, when tested in accordance with 4.5.1.6.

3.5.4 Corrosion resistance. Corrosion resistant metals and corrosion resistant alloys shall be employed in accordance with MIL-STD-454, Requirement 15.

3.5.5 Altitude. The Scoring Set shall be designed to operate when subjected to altitudes up to and including 85,000 feet above sea level for not less than 30 minutes.

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3.5.6 Vibration. The Scoring Set shall meet all requirements of Section 3 when tested in accordance with 4.5.1.5.

3.5.7 Shock. The Scoring Set shall operate as specified herein after being subjected to shock requirements of MIL-STD-810C, Method 516 except that the shock level shall be 15 gravity units (G's).

3.5.8 Interim storage. The Scoring Set shall meet the requirements of this specification after not less than 2.5 years of interim storage (see 6.7).

3.5.9 Long term storage. The Scoring Set shall meet the requirements of this specification after not less than three years of long term storage (see 6.8).

3.5.10 Warm-up time. The Scoring Set shall achieve stabilized operation, within limits as specified herein, under all environmental conditions of paragraph 3.5 in not greater than 60 seconds after the application of operating power.

3.5.11 Salt fog. The Scoring Set shall be designed to withstand exposure to salt fog when tested in accordance with 4.6.1.5.

3.6 Configuration control. Approval from the procuring activity shall be obtained before making any changes in tolerances, dimensions, design, materials, or manufacturing processes subsequent to first article approval.

3.7 Design and construction. Design and construction of the Scoring Set shall be in accordance with Drawing 1356AS3021 and as specified herein.

3.7.1 Microelectronic devices. When used, microelectronic devices shall meet the requirements of MIL-STD-965, Procedure 1.

3.7.1.1 Microelectronic interchangeability. The interchangeability provisions of MIL-STD-965, Procedure 1 shall govern microelectronic interchangeability.

3.7.1.2 Modularization. The electronic portions of the Scoring Set shall be functionally modularized in accordance with MIL-STD-2078.

3.7.1.3 Non-metallic parts and materials. Non-metallic parts and materials shall be non-nutrients to fungi or shall be treated to resist fungus growth and shall meet the requirements of MIL-STD-454, Requirement 4.

3.7.1.3.1 Dissimilar metals. Selection and protection of dissimilar metal combinations shall be in accordance with MIL-STD-454, Requirement 16.

3.8 Finishes. Finishes shall be capable of withstanding a 48-hour salt spray (fog) exposure when tested in accordance with MIL-STD-810C, Method 509.1, Procedure I.

3.8.1 Scoring Set enclosure. Finish of the Scoring Set enclosure shall be lusterless gray color No. 36231 in accordance with FED-STD-595.

3.8.2 Cables and connectors. The Scoring Set shall be designed for the use of cables and connectors in accordance with MIL-E-5400.

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3.8.2.1 Scoring Set enclosure electrical connections. Designation for the external connections on the Scoring Set enclosure shall be labeled in gloss black color No. 17038 in accordance with FED-STD-595 and Table I.

TABLE I. Scoring Set external connection designation.

Connector designation	Connector type	Function
J1	MS 3134E3-5OP	Power
J2	Female SMA	MDI Sensor Antenna
J3	Female TNC	TM Transmitter Antenna
J4	Female BNC	Receiver video output

3.8.2.2 Interconnecting cabling. The Scoring Set shall be designed to operate using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires shall be shielded when demonstrated as necessary to meet the interference control requirements of 3.8.2.3, and provided the assembly of the cable to its plugs shall be easily accomplished.

3.8.2.3 Electromagnetic interference (EMI). The Scoring Set shall not generate conducted or radiated emissions beyond the limits specified in MIL-STD-461B for Class A1 category A1b equipment.

3.9 Nomenclature, nameplates, and identification marking. Nomenclature and serial number assignments shall be in accordance with MIL-N-18307. Identification of assemblies, sub-assemblies, components, and parts shall be in accordance with MIL-STD-130. Nameplates shall be in accordance with Drawing 1356AS3021.

3.9.1 Cable length. The MDI Sensor cable length shall be identified on a removable nameplate or sticker in accordance with Drawing 1356AS3021.

3.9.2 External adjustments. All external adjustments shall be labeled in gloss black color No. 17038 in accordance with FED-STD-595.

3.9.3 Interchangeability. All unique assemblies and parts shall be interchangeable with all like items and shall conform to the requirements of MIL-STD-454, Requirement 7.

3.10 Logistics.

3.10.1 Maintenance design. All SRAs within the Scoring Set shall be readily accessible for removal and fault isolation.

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3.10.2 Special tools. The design of the Scoring Set shall be such that requirements for special tools for tuning, adjusting, or maintenance shall be kept to a minimum. Special tools shall not be required for the installation or replacement of electronic parts. The contractor shall provide the procuring activity with any special tools required for tuning, adjusting, or maintaining the Scoring Set.

3.10.3 Supply. The Scoring Set design shall be logistically supportable by a Navy supply system and shall use, wherever possible, standard National or Naval supply system parts, fittings, hardware, SRAs, etc.

3.11 Soldering. Soldering practices shall be in accordance with MIL-STD-454, Requirement 5.

3.12 Workmanship. Workmanship, fabrication, and assembly practices shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment and facilities. Inspection equipment and facilities shall be of the accuracy, quality, and quantity to permit the performance of the required inspection. The contractor shall establish and maintain adequate calibration of inspection equipment in accordance with MIL-STD-45662.

4.1.2 Subsystem level inspection. There shall be provisions for subsystem level testing when specified in the contract (see 6.2.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. TM Transmitter Acceptance Test (see 4.3)

b. Operating Run-In Test (see 4.4)

c. Production Unit Acceptance Test (see 4.5)

d. First Article Test (see 4.6)

e. Reliability Assurance Test (see 4.7)

f. Special Tests (see 4.8)

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4.2.1 Test procedures. The procedures for conducting the Operating Run-in Test, the Production Unit Acceptance Test, TM Transmitter Acceptance Test, the First Article Test, Reliability Assurance Test, and Special Tests shall be prepared by the contractor and made available to the procuring activity for acceptance (see 6.2.2). The right is reserved for the procuring activity or the Government Representative to modify the tests or require any additional tests necessary to determine compliance with the specification, contract or purchase order. Inspection and test procedures shall be prepared in accordance with MIL-T-18303. Test reports shall conform to MIL-STD-831.

4.2.2 Inspection conditions. Unless otherwise specified herein or in the contract or purchase order (see 6.2.1), quality conformance inspections shall be conducted under the following Standard Atmospheric Conditions:

a. Room temperature - +15°C to +30°C

b. Humidity - Not greater than 98 percent relative

c. Altitude - (barometric pressure) 28 to 31 inches of mercury (Hg)

4.2.3 Accept or reject criteria. The acceptance criteria shall be evidenced by the ability of the Scoring Set to meet the performance requirements within the tolerances specified in Section 3.

4.2.4 Failures. Should a failure (see 6.9) occur during a specific test in the test procedures classified in 4.2, the following actions shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is due to a material defect or a design defect.

c. Prepare for the procuring activity acceptance, the proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests (see 4.2.6 and 6.2.2).

d. Where practical, include a special test to check all Scoring Sets for this requirement until assurance is obtained that the defect has been corrected.

4.2.5 Failure report. In the event of a failure, the Government Representative shall be notified. A report shall be made available to the procuring activity upon completion of test (see 6.2.2).

4.2.6 Failure analysis. Failures shall be subjected to an engineering analysis to determine cause of failure, and corrective action shall be proposed to eliminate failures (see 6.2.2).

4.3 TM Transmitter Acceptance Test. The TM Transmitter Acceptance Test shall be used by the contractor to determine complince with electrical performance and physical characteristic requirements specified herein. This test shall be performed on every TM Transmitter prior to its installation in the Scoring Set. The TM Transmitter Acceptance Test shall consist of the tests of Table II and shall be performed in the sequence specified, therein.

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TABLE II. TM Transmitter Acceptance Test.

Test Requirement	Requirements Paragraph
1. Visual & mechanical inspection	3.4.3.1.3
2. Input voltage & power	
a. undervoltage & current	3.4.1.3.14, 3.4.2.1, 3.4.2.3
b. overvoltage, current & reverse polarity	
c. grounded input	3.4.2.4
d. open and short circuit loads test	3.4.1.3.11
3. Transmitter electrical parameters	
a. output power, frequency, frequency stability, output impedence	3.4.1.3.1, 3.4.1.3.2, 3.4.1.3.5, 3.4.1.3.10
b. spurious emissions, incidental FM, amplitude modulation, modulation frequency response, maximum carrier deviation, deviation sensitivity, 2.0 to 1.0 mismatch test	3.4.1.3.3, 3.4.1.3.5, 3.4.1.3.6, 3.4.1.3.7, 3.4.1.3.8, 3.4.1.3.9 3.4.1.3.12
4. Environmental tests	
a. low and high temperature	3.5.1.2

4.4. Operating Run-In Test. The Operating Run-In Test shall be used by the contractor to eliminate infant mortality prior to performing the First Article, Reliability Assurance, and Production Unit Acceptance Tests. Unless otherwise specified in the contract, this test shall be performed on every Scoring Set (see 6.2.1). The Operating Run-In Test shall consist of ambient temperature burn-in as specified below:

a. Ambient temperature burn-in test shall be conducted to verify that the Scoring Set is operational before it is subjected to any temperature

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conditions. During this test the following parameters shall be monitored: (1) Sensor transmitter frequency and power output, (2) TM Transmitter frequency and power output, (3) Loop sensitivity at 75 feet.

b. Eight-hour ambient burn-in test at 32 VDC shall be conducted by running two four-hour cycles at ambient burn-in with power applied to the Scoring Set during the entire eight-hour period. After each cycle, the same parameter shall be monitored as specified in 4.4a.

4.5 Production Unit Acceptance Test. The Production Unit Acceptance Test shall be used by the contractor to determine compliance with the requirements specified herein for material, workmanship, watertight integrity, and compliance with the performance, electrical, and physical characteristic requirements specified herein. Unless otherwise specified in the contract or purchase order (see 6.2.1), this test shall be performed on every Scoring Set. The Production Unit Acceptance Test shall consist of the tests of Table III and shall be performed in the sequence specified in the contract or purchase order (see 6.2.1).

4.5.1 Production Unit Acceptance Test method. Production Unit Acceptance Test method shall be as specified in 4.5.1.1 through 4.5.1.6.

4.5.1.1 Visual and mechanical inspection. The contractor shall inspect each Scoring Set to determine that it meets all requirements for workmanship, identification, marking, finish, weight, dimensional accuracy, cabling and connectors as specified herein.

4.5.1.2 Input voltage and power. The contractor shall conduct this test to demonstrate that all input voltage and power requirements specified in Table III have been met.

4.5.1.3 Sensor electrical parameters. The contractor shall conduct this test to demonstrate that all electrical parameters requirements specified in Table III have been met.

4.5.1.4 TM Transmitter and VCO. The contractor shall conduct this test to demonstrate that all TM Transmitter and VCO requirements specified in Table III have been met.

4.5.1.5 Random vibration test. The contractor shall vibrate each Scoring Set in accordance with MIL-STD-810C, Method 514.2, Table 514.2-V, Figure 514.1-5 Curve AE, without vibration isolators for three minutes in each axis. The Scoring Set MDI Sensor antenna Jack (J2) shall be connected to 50 ohms and 1.5 to 1.0 VSWR load via the length of RG-142 B/U cable (12.5 feet) for which the system was set up. The receiver video output shall be monitored on an oscilloscope with a sweep setting of five or ten milliseconds per centimeter and the TM Transmitter output shall be monitored on a spectrum analyzer for spurious emissions, incidental FM, and amplitude modulation (AM). The receiver video noise output shall be not greater than two times the receiver video output average noise spike level, and shall be not greater than six dB increase on an rms voltmeter.

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TABLE III. Production Unit Acceptance Test.

Test requirement	Requirements paragraph	Test method paragraph
1. Visual and mechanical inspection	3.4.3	4.5.1.1
2. Input voltage and power	3.4.2.1	4.5.1.2
a. Undervoltage and current b. Overvoltage and current, and reverse polarity test for 30 seconds	3.4.2.1, 3.4.2.3	
c. Grounded input	3.4.2.4	
d. No load conditon	3.4.2.5	
e. Degraded performance test	3.4.2.2	
f. Open and short circuit loads test for 60 seconds for TM and sensor XMTRs	3.4.1.1.18, 3.4.1.3.11	
3. Sensor electrical parameters		4.5.1.3
a. Transmitter output power and frequency	3.4.1.1.5, 3.4.1.1.6	
b. Pulse width, pulse rise and fall time, PRF, tilt test	3.4.1.1.7, 3.4.1.1.8	
c. Receiver sensitivity test	3.4.1.1.9	
d. Loop sensitivity, doppler output, frequency response, and internal sensor timing test	3.4.1.1.11, 3.4.1.1.12, 3.4.1.1.13	
e. Receiver video output noise, internally-generated nonscoring signals, and cable phase sensitivity test	3.4.1.1.10, 3.4.1.1.16, 3.4.1.1.19	
4. TM Transmitter and VCO		4.5.1.4
a. Transmitter, output power, frequency and deviation, spurious emissions	3.4.1.3.1, 3.4.1.3.3 3.4.1.3.5, 3.4.1.3.6	
b. Voltage Controlled Oscillator center frequency, deviation, output voltage and linearity	3.4.1.2.1, 3.4.1.2.2, 3.4.1.2.3	
c. Rejection of telemetry signals	3.4.1.1.15	
5. Random vibration test	3.4.1.1.17, 3.5.6	4.5.1.5
6. Enclosure test	3.5.3	4.5.1.6

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4.5.1.6 Enclosure test. The Scoring Set shall be subjected to an enclosure test in accordance with MIL-STD-810C, Method 512.1, Procedure IV. Any evidence of leakage shall be not acceptable.

4.5.2 Data sheets. Test data sheets shall be included with every Scoring Set showing all Production Unit Acceptance Test results (see 6.2.2).

4.5.3 Adjustments. If adjustments or component changes are required for the selected Scoring Set to pass a particular test, all previous tests shall be repeated unless an exception is granted by the Government representative.

4.5.4 Rejection and retest. Before resubmitting a previously rejected Scoring Set to procuring activity, the Production Unit Acceptance Test shall be rerun and new data sheets, along with a failure report, shall be included with the Scoring Set.

4.6 First Article Test. Before this test is run, the first article shall undergo the Operating Run-In Test. The First Article Test shall consist of the Production Unit Acceptance Test and tests to determine that the Scoring Set meets all of the Scoring Set environmental and EMI requirements specified herein. The First Article Test shall consist of the test of Table IV and shall be performed in the sequence listed.

TABLE IV. First article acceptance test.

Test requirement	Requirements paragraph	Test method paragraph
1. Operating run-in tests		4.4
2. Production Unit Acceptance Test except vibration test shall be run for 30 minutes in each axis (See Table III)	See Table III	See Table III
3. Environmental tests		
a. Low and high temperature test	3.5.1.2	4.6.1.1
b. Temperature altitude test	3.5.1.2, 3.5.5	4.6.1.2
c. Shock test	3.5.7	4.6.1.3
d. Humidity test	3.5.2	4.6.1.4
e. Salt fog test	3.5.11	4.6.1.5
4. Electromagnetic interference tests per MIL-STD-462 CE03, CE06	3.8.2.3	4.6.2.1

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4.6.1 Environmental tests. Environmental tests shall be conducted to verify that the Scoring Set meets the requirements of Section 3. Test data shall be collected before, during, and after the test environment, with no degradation allowable from the levels specified in 3.5.

4.6.1.1 Low and high temperature test. The Scoring Set shall be tested in accordance with MIL-STD-810C, Method 501.1, Procedure I and Method 502.1, to the temperature limits specified in 3.5.

4.6.1.2 Temperature-altitude test. The Scoring Set shall be tested in accordance with MIL-STD-810C, Method 504.1, Procedure I, Equipment catagory 7, the temperature and altitude limits specified in 3.5.

4.6.1.3 Shock test. The Scoring Set shall be tested in accordance with MIL-STD-810C, Method 516.2, Procedure I, Figure 516.2-1, 15 G's peak amplitude for 11 milliseconds duration. The Scoring Set shall be subjected to three shocks per direction per axis for a total of 18 shocks.

4.6.1.4 Humidity test. The Scoring Set shall be tested in accordance with MIL-STD-810C, Method 507.1, Procedure I.

4.6.1.5 Salt fog test. The Scoring Set shall be tested in accordance with MIL-STD-810C, Method 509.1, Procedure I. Test data shall be collected prior to exposure, again within one hour after exposure, and again at the end of the drying period.

4.6.2 Electromagnetic interference.

4.6.2.1 Emissions. The Scoring Set shall be tested as specified in MIL-STD-462, CE03, and CE06 for Class A1 category A1b equipment.

4.6.3 Adjustments. If adjustments or component changes are required for the selected Scoring Set to pass a particular test, all previous tests shall be repeated.

4.6.4 First article acceptance test data. The contractor shall prepare a first article test report including all data collected in conducting these tests (see 6.2.2).

4.6.5 Approval of the first article. The first article sample will be approved by the procuring activity after successful completion of first article inspection. No production Scoring Sets will be accepted prior to approval of the first article sample. Fabrication prior to approval of the first article sample shall be at the risk of the contractor.

4.7 Reliability Assurance Test. A Reliability Assurance Test shall be conducted in accordance with MIL-STD-781. The production acceptance (sampling) phase shall be in accordance with MIL-STD-781, Test Plan XIV-C (Chart 15) and MIL-STD-2074. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventative maintenance to be allowed during the test and so forth, shall be part of the test procedures to be

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prepared prior to the beginning of the reliability assurance tests (see 4.2.1). As a minimum, the following parameters shall be monitored during reliability tests: (1) Sensor transmitter frequency and power output, (2) TM Transmitter frequency and power output, (3) Loop sensitivity at 75 feet.

4.7.1 All Scoring Set production reliability acceptance test. Each Scoring Set produced shall be tested for ten failure-free hours of operating time. Prior to the ten hour failure-free test, a burn-in period may be used. If the burn-in period is used, the details thereof shall be included in the procedures. To determine whether the MTBF requirement specified herein is being met at any time during the contract, the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the accept-reject criteria of Figure C-13 of MIL-STD-781. These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. The Government representative shall be notified any time that the current totals of test hours and test failures plotted on Figure C-13 of MIL-STD-781 show a reject situation (see 4.2.7). The procuring activity reserves the right to stop the acceptance of the Scoring Set at any time that a reject situation exists pending a review of the contractor's efforts to improve the Scoring Set, the Scoring Set parts, the Scoring Set workmanship, etc., so that the entire compilation will show other than a reject situation. The test levels shall be as follows:

a.	Temperature (cycling):	-40 to +71°C
b.	Vibration:	See 4.5.1.5
c.	Operating power:	+24 to +32 VDC
d.	Scoring Set on-off cycling:	Scoring Set off during cooling cycle and on during heat cycle

4.8 Special Tests. Unless otherwise specified in the contract or purchase order (see 6.2.1), Special Tests shall be conducted for the purpose of checking the effect of a design or material change on the performance of the Scoring Set and to assure quality control. The Scoring Set selected for Special Tests may be selected from Scoring Sets previously subjected to First Article or Production Unit Acceptance Tests.

4.8.1 Special Test schedule. Selection of Scoring Sets for Special Tests shall be made as follows:

a. On a Scoring Set selected by the procuring activity after an engineering or material change

b. Whenever failure reports or other information indicate to the procuring activity that additional tests are required

4.8.2 Scope of tests. Special Tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the First Article or Production Unit Acceptance tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and make it available to the procuring activity for acceptance prior to conducting the tests (see 6.2.2).

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4.9 In-process inspection and verification.

4.9.1 Design and construction. To determine conformance to the drawing requirements of 3.4.3.1 and 3.7, the contractor shall perform the necessary visual and dimensional examinations to verify conformance to the dimensional and marking requirements of Drawing 1356AS3021.

4.9.2 Materials, parts, and processes verification. Materials, parts, and processes as specified herein will be verified by a cognizant Government representative as follows:

a. Evidence of satisfactory completion of the examination and analysis.

b. Quality assurance documents showing that parts, materials, and processes applied at the critical item assembly level have been checked and accepted by a cognizant Government representative.

c. Documents showing inspection and acceptance of all items at the next lower assembly level to assure that parts, materials, and processes meet the requirements specified herein.

4.9.3 Corrosion resistance. Inspections shall be made visually or by documentation that corrosion resistant metals and corrosion resistant alloys have been utilized in accordance with MIL-STD-454, Requirement 15.

4.9.4 Nameplate and product marking. Verification for nameplate and product marking requirements specified herein shall be made by visual examination, using the applicable inspection procedure. This verification shall be applied to all nameplates and marking added during the critical item assembly operation. Additional visual examination shall be required on lower level assemblies to assure that damage to nameplates or marking has not occurred. This verification shall also be required for exterior or high visibility items. Otherwise, verification of lower assembly level items shall be by reference to documents showing inspection and acceptance at the next level of assembly. All assemblies shall be identified by a Scoring Set serial number for traceability of configuration.

4.9.5 Interchangeability. Interchangeability of like replaceable items will be verified if an examination and analysis confirms that they possess the following characteristics:

a. Equal in performance

b. Equal in durability

c. Capable of being interchanged without alteration to the item

4.9.6 Safety. The Scoring Set will be considered verified for safety after evidence of satisfactory completion of acceptance tests as specified herein.

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4.9.7 Human performance and human engineering. The Scoring Set will be considered verified for human performance and human engineering after satisfactory completion of tests specified herein.

4.9.8 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-E-17555 and MIL-STD-794.

4.9.9 Soldering. Verification shall be made that soldering conforms to the requirements of MIL-STD-454, Requirement 5.

4.9.10 Workmanship. Each Scoring Set will be examined by a cognizant Government representative to ensure that parts and materials fabricated, produced, or purchased conform to MIL-STD-454, Requirement 9.

5. PACKAGING

5.1 Preservation. Each Scoring Set shall be preserved and packaged in accordance with MIL-E-17555 and MIL-STD-794 for the following applicable level or levels as specified in the contract or purchase order (see 6.2.1).

5.1.1 Level A. Adequate protection against corrosion, deterioration, and physical damage during multiple reshipment, handling and intermediate storage, and worldwide redistribution.

5.1.2 Level B. Adequate protection against corrosion, deterioration, and physical damage during multiple reshipment, handling and known storage conditions for periods normally not greater than one year.

5.1.3 Level C. Adequate protection against corrosion, deterioration, and damage during shipment from supply source to the first receiving activity for immediate use (vendor to user).

5.2 Packing. Each Scoring Set shall be packed in accordance with MIL-E-17555 and MIL-STD-794 for the following applicable level or levels as specified in the contract or purchase order (see 6.2.1).

5.2.1 Level A. Packing which shall afford adequate protection against damage during shipment, handling, indeterminate storage, and worldwide distribution.

5.2.2 Level B. Packing which shall afford adequate protection against multiple domestic shipments, multiple handling, and covered storage.

5.2.3 Level C. Packing which shall afford adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.3 Marking. Marking of individual unit and intermediate packages and exterior shipping containers shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The Scoring Set is intended for use as a Miss Distance Indicator for aerial and surface targets. The Scoring Set will be used in conjunction with Naval Range telemetry receiving stations and may also be used with suitable shipboard portable receiving sets currently under development. The Scoring Set will be used in both missile and gunnery type operations. The received data will be reduced to yield miss distance information in the missile firings. In ship's gunnery type firings, automatic scoring of the number of projectiles within a specified scoring zone can be accomplished.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Quantity of Scoring Sets to be procured.

c. First article sample (see 3.1.1).

d. First article inspection if other than specified (see 3.3.1 and 4.6).

e. Responsibility for inspection if different from 4.1.

f. Provisions for subsystem level testing, if required (see 4.1.2).

g. Inspection conditions if other than as specified in 4.2.2.

h. Operating Run-In Test if other than as specified in 4.4.

i. Production Unit Acceptance Test if other than as specified in 4.5.

j. Sequence for performing Production Unit Acceptance Test (see 4.5).

k. Special Tests designated (see 4.8).

l. Specific level of preservation, packaging, and packing (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR-7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

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Paragraph no.	Data requirement title	Applicable DID no.	Option
3.2.1	Nonstandard Part Approval Request/Proposed Additions to an Approved Program Parts Selection List	DI-E-7028A	---
3.4.4	Reports, Reliability Test and Demonstration	DI-R-7034	---
4.2.1, 4.8.2	Procedures, Test	UDI-T-21347	---
4.2.5, 4.2.6	Failure Analysis and Corrective Action Report	DI-R-5299C	---
4.5.2	Records, Inspection, and Test	UDI-T-22706A	---
4.6.4	Report, First Article (Preproduction) Test	UDI-T-21349	---

(Data item descriptions related to this specification, and identified in Section 6, will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractor in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one Scoring Set. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, approval, and disposition of the first article sample.

6.4 Definition of a monotonically increasing function. A monotonically increasing function is a mathematical function that never decreases as the independent variable increases.

6.5 Definition of the loop sensitivity of the Scoring Set at any given range and frequency. The loop sensitivity of the Scoring Set at any given range and frequency is the amount of attenuation required to produce a signal at the sensor receiver video output jack (J4) with a six dB signal plus noise-to-noise ratio.

6.6 Definition of baseplate. The baseplate is the mounting plate of the Scoring Set.

6.7 Definition of interim storage. Other than when installed in a target, or when in long term storage, the Scoring Set is considered in interim storage at all times following delivery to the procuring activity. Interim storage includes the period of factory or depot maintenance, factory or depot storage,

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and land, sea, and air transportation. The relative humidity during interim storage is not greater than 90 percent at temperatures from zero to 51°C, or greater than 30 percent at temperatures from -29 to 0°C for periods greater than 24 hours.

6.8 Definition of long term storage. The Scoring Set is considered in long term storage when packaged and packed in accordance with the requirements of 5.1. The relative humidity within the container during long term storage will be not greater than 15 percent at temperatures from -37 to -29°C, not greater than 30 percent at temperatures from -29 to 0°C, and not greater than 50 percent at temperatures from zero to 71°C for periods not less than 24 hours.

6.9 Definition of failure. A failure is an out-of-specification condition as specified herein.

Preparing activity
Navy - AS
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-85688A	Scoring Set, Missile, Non-Cooperative System AN/DSQ-37(V)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER (Specify): _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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